

Estimated Average Requirements (EAR)

The Estimated Average Requirement (EAR) is the average daily nutrient intake level estimated to meet the requirements of half the healthy individuals in a particular life stage and gender group. It serves as a reference point to assess the adequacy of nutrient intake within a population. EAR is determined through scientific research and is used as a basis for setting the Recommended Dietary Allowances (RDAs).

Applications of EAR in the field of nutrition, health and public policy:

- **Research and Population Assessment:** EAR is primarily used in scientific research to assess the nutritional status of populations. It helps researchers understand the average nutrient needs within specific groups based on age, gender, and life stage.
- **Setting Dietary Reference Intakes (DRIs):** EAR serves as the foundation for establishing other components of DRIs, such as Adequate Intakes (AI) and Tolerable Upper Intake Levels (UL). DRIs provide a comprehensive framework for dietary recommendations.
- **Public Health Surveys:** EAR values are utilized in national and international surveys to evaluate the adequacy of nutrient intake within populations. These surveys help identify trends and potential nutritional deficiencies.

Summary of Estimated Average Requirements (EAR) of Nutrients for Indians																			
Age Group	Category of ...	Referenc...	Energy (Kcal/Kg/d)	Dietary Fibre (g/d)	Protein (g/Kg/d)	VitaminA (µg/d)	Thiamin (mg/d)	Riboflavin (mg/d)	Niacin (mg/d)	VitaminC (mg/d)	Vitamin-B6 (mg/d)	Folate (µg/d)	Vitamin-B12 (µg/d)	VitaminD (IU/d)	Calcium (mg/d)	Magnesium (mg/d)	Iron (mg/d)	Zinc (mg/d)	Iodine (µg/d)
Infants	6-12 Months	8.5	80	-	1.04	170	-	-	-	-	0.5	71	1	-	-	-	2	2.1	-
Children	1-3 Years	12.9	83	-	0.79	180	0.6	0.8	6	24	0.8	97	1	400	400	73	6	2.8	65
Children	4-6 Years	18.3	74	-	0.7	240	0.8	1.1	8	27	1	111	2	400	450	104	8	3.7	65
Children	7-9 Years	25.3	67	-	0.75	290	1	1.3	10	36	1.3	142	2	400	500	144	10	4.9	65
Boys	10-12 Years	34.9	64	-	0.75	360	1.3	1.7	12	45	1.7	180	2	400	650	199	12	7	70
Boys	13-15 Years	50.5	57	-	0.72	430	1.6	2.2	16	60	2.2	238	2	400	800	287	15	11.9	100
Boys	16-18 Years	64.4	52	-	0.7	480	1.9	2.1	19	70	2.5	286	2	400	850	367	18	14.7	100
Girls	10-12 Years	36.4	57	-	0.73	370	1.2	1.6	12	44	1.6	186	2	400	650	207	16	7.1	70
Girls	13-15 Years	49.6	49	-	0.7	420	1.3	1.9	13	55	1.8	204	2	400	800	282	17	10.7	100
Girls	16-18 Years	55.7	45	-	0.67	400	1.4	1.9	14	57	1.9	223	2	400	850	317	18	11.8	100
Men	Sedentary	65	32	-	0.66	460	1.2	1.6	12	65	1.6	250	2	400	800	370	11	14.1	95
Men	Moderate	65	42	-	0.66	460	1.5	2.1	15	65	2.1	250	2	400	800	370	11	14.1	95
Men	Heavy	65	53	-	0.66	460	1.9	2.7	19	65	2.6	250	2	400	800	370	11	14.1	95
Women	Sedentary	55	30	-	0.66	390	1.1	1.6	9	55	1.6	180	2	400	800	310	15	11	95
Women	Moderate	55	39	-	0.66	390	1.4	2	12	55	1.6	180	2	400	800	310	15	11	95
Women	Heavy	55	49	-	0.66	390	1.8	2.6	15	55	2.1	180	2	400	800	310	15	11	95
Women	Pregnant W...	65	30	-	ester), 0.66 +	406	1.6	2.3	11	65	1.9	480	2.2	400	800	370	21	12	160
Women	Lactation(0-...	39	-	-	13.6 (2nd trin	720	1.7	2.5	13	95	1.82	280	2.8	400	1000	335	16	11.8	200
Women	Lactation(7-...	39	-	-	10.6 (3rd trin	720	1.7	2.4	13	95	1.76	280	2.8	400	1000	335	16	11.8	200
Elderly	Men >= 60 ...		1700	-	43	460	1.2	1.6	12	65	1.6	250	2	400	1000	370	11	14	95
Elderly	Women >= ...		1500	-	36.3	390	1.1	1.6	9	55	1.6	180	2	400	1000	310	11	11	95

2. Recommended Dietary Allowances (RDA):

Recommended Dietary Allowances (RDAs) are the average daily dietary nutrient intake levels sufficient to meet the nutrient requirements of nearly all (97-98%) healthy individuals in a specific life stage and gender group. RDAs are set based on the EAR, taking into account individual variability. RDAs are designed to prevent nutrient deficiencies and maintain optimal health.

Applications of RDA in the field of nutrition, health and public policy:

- **Individual Dietary Planning:** RDA is commonly used by nutritionists and dietitians when developing personalized dietary plans for individuals. It provides a target for nutrient intake that is sufficient to meet the needs of nearly all individuals in a specific life stage and gender.
- **Clinical Practice:** Healthcare professionals use RDAs to assess and manage the nutritional needs of patients. RDAs guide interventions to prevent or address nutrient deficiencies and promote overall health.
- **Food Labeling:** RDA values are utilized in the development of nutrition labels on packaged foods. These labels help consumers understand the percentage of the recommended daily intake provided by a particular food item.

- Public Health Guidelines: RDAs contribute to the establishment of public health guidelines and policies related to nutrition. Government agencies use these values to inform recommendations for healthy eating habits at the population level.

Summary of Recommended Dietary Allowances (RDA) of Nutrients for Indians																				
Age Group	Category of ...	Reference body weight (Kg)	Energy (Kcal/Kg/d)	Dietary Fibre (g/d)	Protein (g/Kg/d)	VitaminA (µg/d)	Thiamin (mg/d)	Riboflavin (mg/d)	Niacin (mg/d)	VitaminC (mg/d)	Vitamin-B6 (mg/d)	Folate (µg/d)	Vitamin- B12 (µg/d)	VitaminD (IU/d)	Calcium (mg/d)	Magnesium (mg/d)	Iron (mg/d)	Zinc (mg/d)	Iodine (µg/d)	
Infants	6-12 Months	8.5	-	-	1.23	350	0.4	0.6	5	30	0.6	85	1.2	400	300	75	3	2.5	130	
Children	1-3 Years	12.9	-	15	0.97	390	0.7	1.1	7	30	0.9	120	1.2	600	500	90	8	3.3	90	
Children	4-6 Years	18.3	-	20	0.87	510	0.9	1.3	9	35	1.2	135	2.2	600	550	125	11	4.5	90	
Children	7-9 Years	25.3	-	26	0.92	630	1.1	1.6	11	45	1.5	170	2.2	600	650	175	15	5.9	90	
Boys	10-12 Years	34.9	-	33	0.91	770	1.5	2.5	15	55	2	220	2.2	600	850	240	16	8.5	100	
Boys	13-15 Years	50.5	-	43	0.89	930	1.9	2.7	19	70	2.6	285	2.2	600	1000	345	22	14.3	140	
Boys	16-18 Years	64.4	-	50	0.86	1000	2.2	3.1	22	85	3	340	2.2	600	1050	440	26	17.6	140	
Girls	10-12 Years	36.4	-	30	0.9	790	1.4	1.9	14	50	1.9	225	2.2	600	850	250	28	8.5	100	
Girls	13-15 Years	49.6	-	36	0.87	890	1.6	2.2	16	65	2.2	245	2.2	600	1000	340	30	12.8	140	
Girls	16-18 Years	55.7	-	38	0.83	860	1.7	2.3	17	70	2.3	270	2.2	600	1050	380	32	14.2	140	
Men	Sedentary	65	-	30	0.83	1000	1.4	2	14	80	1.9	300	2.2	600	1000	440	19	17	140	
Men	Moderate	65	-	40	0.83	1000	1.8	2.5	18	80	2.4	300	2.2	600	1000	440	19	17	140	
Men	Heavy	65	-	50	0.83	1000	2.3	3.2	23	80	3.1	300	2.2	600	1000	440	19	17	140	
Women	Sedentary	55	-	25	0.83	840	1.4	1.9	11	65	1.9	220	2.2	600	1000	370	29	13.2	140	
Women	Moderate	55	-	30	0.83	840	1.7	2.4	14	65	1.9	220	2.2	600	1000	370	29	13.2	140	
Women	Heavy	55	-	40	0.83	840	2.2	3.1	18	65	2.4	220	2.2	600	1000	370	29	13.2	140	
Women	Pregnant W...	65	-	-	nester), 0.83	900	2	2.7	13	80	2.3	570	2.45	600	1000	440	27	14.5	220	
Women	Lactation(0...	-	-	-	+ 17 (2nd trim	950	2.1	3	16	115	2.16	330	3.2	600	1200	400	23	14.1	280	
Women	Lactation(7...	-	-	-	+ 13 (3rd trim	950	2.1	2.9	16	115	2.07	330	3.2	600	1200	400	23	14.1	280	
Elderly	Men >= 60 ...	-	30	54	1000	1.4	2	14	80	1.9	300	2.2	800	1200	440	19	17	140		
Elderly	Women >= ...	-	25	46	840	1.4	1.9	11	65	1.9	200	2.2	800	1200	370	19	13.2	140		

Note:

1. EAR is more focused on research and assessing the nutritional status of populations.
2. RDA is commonly applied in individual dietary planning, clinical settings and "supervised" nutrition counselling.
3. RDA and EAR both play essential role in ensuring adequate nutrition to support optimal health to individuals and population respectively.